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Understanding Linear Regression

Unlocking Predictive Insights

Introduction

Regression

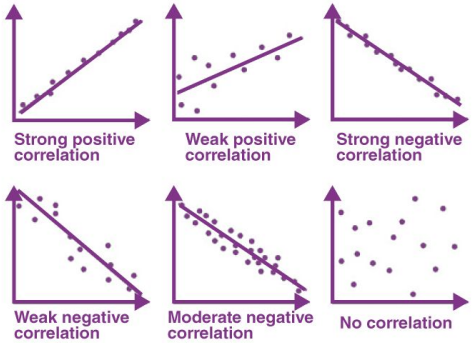
Regression is a statistical method used in various fields to understand and analyze the **relationship** between one or more independent variables and a dependent variable.

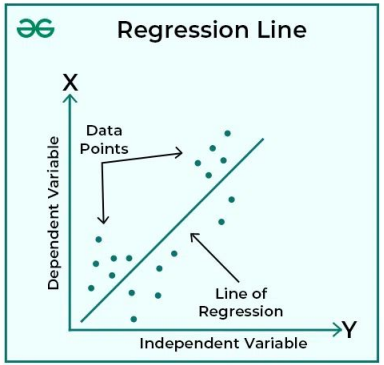
Its primary goal is to predict the value of the dependent variable based on the values of the independent variables.

Regression Vs Correlation

Correlation: Measures the strength and direction of the relationship between two quantitative variables. Does not imply causation, only association. Provides a correlation coefficient (e.g., Pearson's r) ranging from -1 to 1.

Regression: Models the relationship between a dependent variable and one or more independent variables to predict the value of the dependent variable. Focuses on prediction. *It estimates the parameters of regression equation*

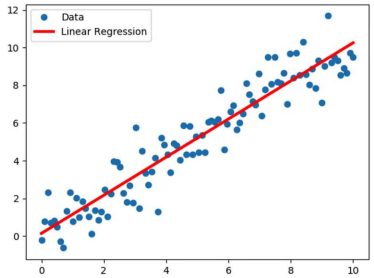




Types of Regression

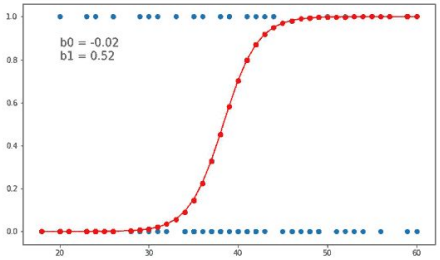
Linear Regression

It assumes a linear relationship between the independent and dependent variables.



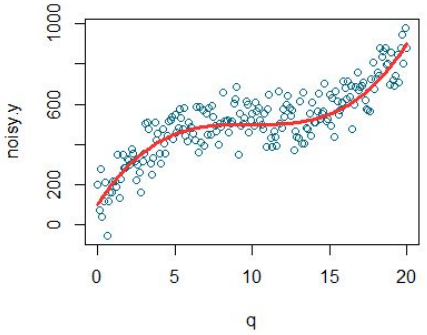
Logistic Regression

It predicts the probability of occurrence of an event by fitting data to a logistic function.



Polynomial Regression

It fits a curve to the data using a polynomial equation.



Linear Regression

What is Linear Regression

Linear regression is a statistical method used to analyze the relationship between two continuous variables.

It assumes a linear relationship between the independent and dependent variables. The equation of a simple linear regression model can be represented as:

Linear Function

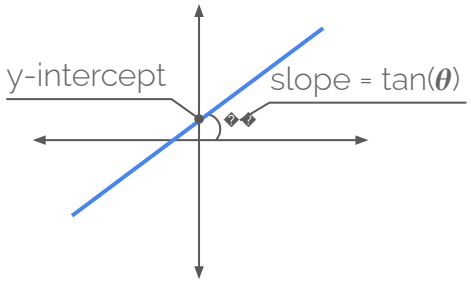
$$y = mx + c \quad \text{or}$$

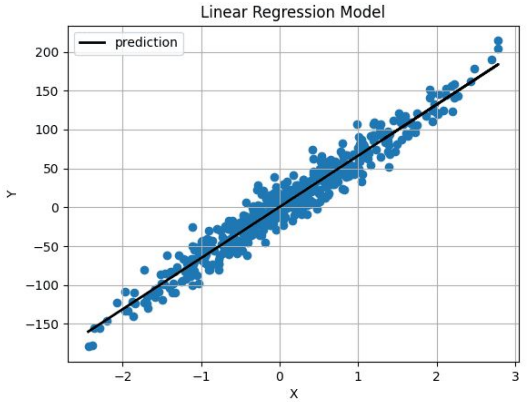
$$h_\theta(x) = \theta_0 + \theta_1 x$$

where,

$\theta_1 = m$ is slope

$\theta_0 = c$ is y-intercept





Types of Linear Regression

Simple Linear Regression

In simple linear regression, there is only one independent variable predicting the dependent variable. The relationship between the variables can be represented by a straight line equation:

$$y = mx + c$$

Where y is the dependent variable, x is the independent variable, m is the slope of the line (the change in y for a unit change in x), and c is the intercept (the value of y when x is 0).

Multiple Linear Regression

In multiple linear regression, there are two or more independent variables predicting the dependent variable. The relationship is represented by the equation:

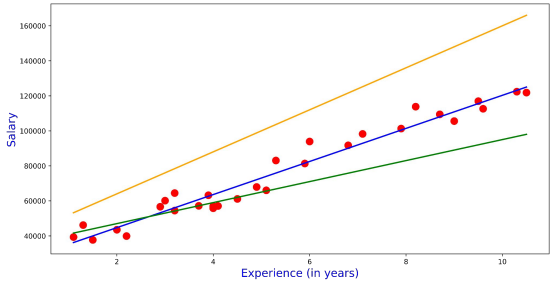
$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

where y is the dependent variable, $x_1, x_2, x_3, \ldots, x_n$ are the independent variables, b_0 is the intercept and $b_1, b_2, b_3, \ldots, b_n$ are the coefficients of the independent variables.

Steps to Perform Linear Regression

1. **Data Collection**
   Gather data on the variables of interest.
2. **Data Preprocessing**
   Clean the data, handle missing values, and perform feature scaling if necessary.
3. **Split Data**
   Divide the data into training and testing sets.
4. **Model Building**
   Fit the linear regression model to the training data.
5. **Model Evaluation**
   Evaluate the model's performance using metrics like Mean Squared Error (MSE), R-squared, etc.
6. **Prediction**
   Use the model to make predictions on new data.

Training of Linear Regression Model



Model 1

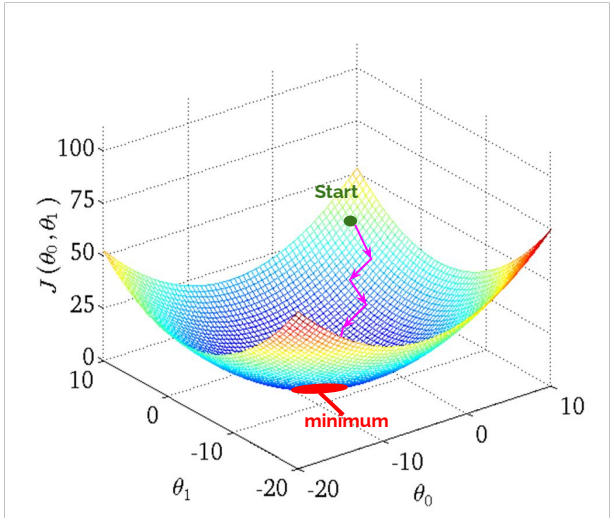
`f(x) = 12000*x+40000`

Model 2

`f(x) = 6000*x+35000`

Model 3

`f(x) = 9449*x+25792`



Gradient Descent

repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}

$Hypothesis : h_\theta(x) = \theta_0 + \theta_1 x$

$Parameters : \theta_0, \theta_1$

$CostFunction : J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$

$Goal : \min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$

Applications of Linear Regression

1. **Economics**
   Linear regression is extensively used in economics to analyze relationships between economic variables such as GDP, inflation, unemployment rate, and consumer spending. It helps economists forecast economic trends and evaluate policy impacts.

2. **Sports Analytics**
   Linear regression is utilized in sports analytics to analyze player performance, predict game outcomes, and optimize team strategies. It helps coaches and team managers make data-driven decisions in areas such as player selection, training regimens, and game tactics.

3. **Medicine and Healthcare**
   Linear regression is used in medical research to study the relationship between variables such as drug dosage and patient outcomes, or between risk factors and disease occurrence. It aids in predicting patient outcomes, understanding disease progression, and assessing the effectiveness of treatments.



Thanks

Any questions?

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